

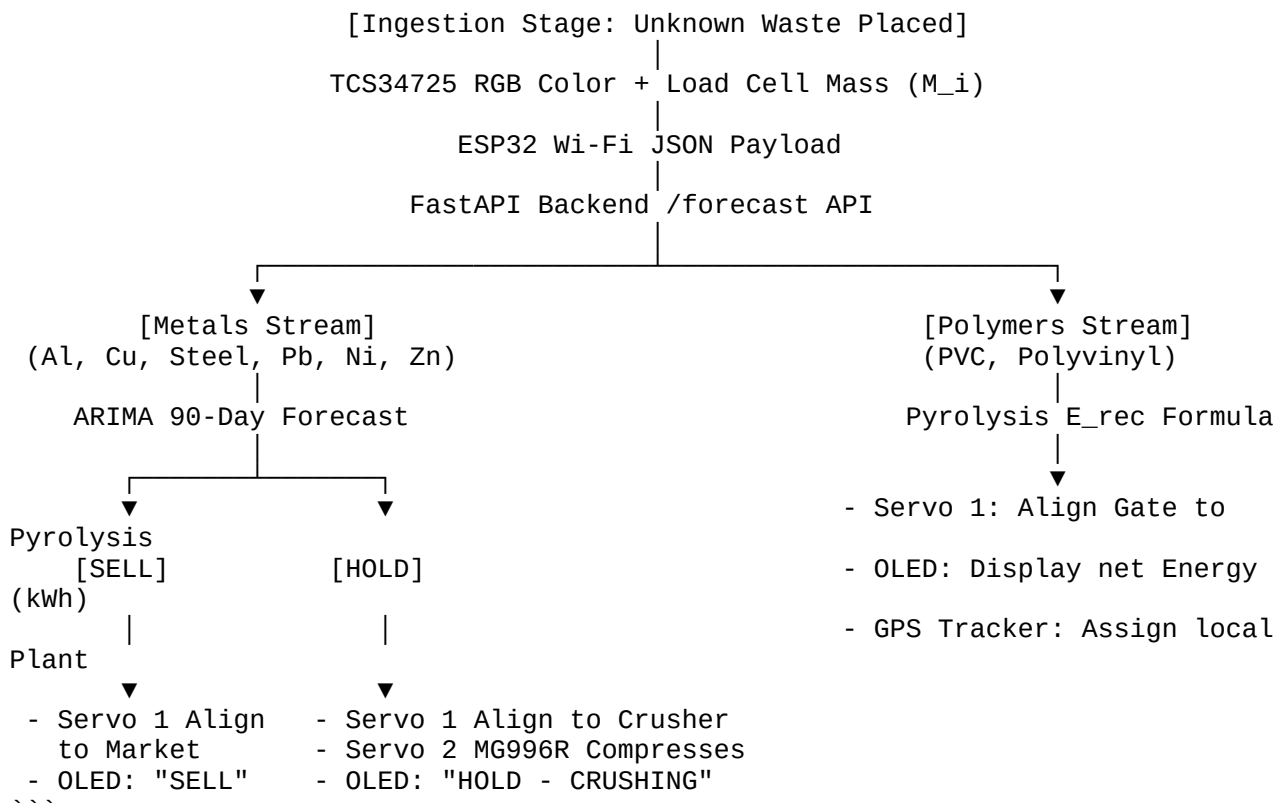
This comprehensive, integrated conceptual design and fully updated IoT hardware list connects **both halves of your component**—**Predictive Metal Valuation** AND the **Strategic Disposition & Energy Recovery Pipeline**—into a highly cohesive, academic, and budget-friendly research prototype [cite: 1, 16].

By replacing the expensive TI LDC1612 with the **TCS34725 RGB Color & Reflectivity Sensor**, this design uses advanced **Sensor Fusion (Weight + Optical)** to identify materials, simulate a physical crushing mechanism, route residues, and execute safety lockout triggers [cite: 16, 117].

1. The Integrated Concept: Closed-Loop Smart Ingestion, Compaction & Logistical Disposition Terminal

This prototype physically simulates how your software models behave when a piece of waste is introduced. It translates your abstract Python machine learning and thermodynamic models into concrete, physical actions that your evaluation panel can easily observe and score [cite: 117, 131].

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Stage A: Dual-Verification Ingestion (Sensor Fusion)

- Upstream Coordination (Component 1):** The overarching system identifies general object profiles (e.g., "cable wires" or "unlabeled plastics") [cite: 104].
- Local Validation (TCS34725 RGB Sensor):** The **TCS34725** is positioned looking down at the scale bed [cite: 128]. It shines its white LED [cite: 128]. When an object is placed, the "Clear/Lux" intensity drops drastically, acting as a contactless **Material Presence Proximity Detector** to trigger the ESP32.
- Mass Ingestion (M_i):** The **Load Cell + HX711** measures the material mass in kilograms [cite: 128].
- The Wi-Fi Dispatch:** The ESP32 packages the optical reflectance channels `[R, G, B, Lux]` and weight `[kg]` into a JSON payload and transmits it over Wi-Fi to your FastAPI backend [cite: 129, 130].

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#### **Stage B: Software Processing & Actuator Decisions**
* **The Predictive Valuation Path (For Base Metals):**
* If the backend classifies the item as an industrial base metal
(Aluminium, Copper, Steel, Lead, Nickel, or Zinc), it triggers your 90-day
**ARIMA/LSTM forecasting model** [cite: 10, 135].
* **"SELL NOW" (Green LED):** If prices are peaking, **Servo 1 (Routing
Gate)** swings to direct the scrap into the market liquidation bin [cite: 3,
46].
* **"HOLD" (Yellow LED + Compactor):** If scrap values are low, the engine
issues a "HOLD" decision [cite: 46]. **Servo 1 (Routing Gate)** swings to guide
the piece to the compaction bay. **Servo 2 (MG996R High-Torque Crusher)** is
then activated [cite: 46]. It rotates a mechanical press arm downward,
physically compacting/crushing the metal on the platform to prepare it as space-
saving feedstock blocks for bulk secondary remake pipelines [cite: 46].
* **The Strategic Disposition Path (For Polymers):**
* If the material is a non-recyclable polymer (like PVC or Grade C
plastics), the backend bypasses the financial forecasting engine and routes it
directly to pyrolysis [cite: 11, 12].
* **Thermodynamic Calculations:** The backend runs your thermodynamic
equation [cite: 12]:

$$\eta = \frac{M_i \times LHV_i}{E_{rec}}$$

* (Using  $LHV_{PVC} = 20.0 \text{ MJ/kg}$ ) and conversion
efficiency  $\eta = 67\%$  to compute the net energy recovery potential in
kWh [cite: 4, 182].
* **Logistical Routing & Display:** **Servo 1 (Routing Gate)** rotates to
direct the plastic into the Pyrolysis Feedstock Bin [cite: 11, 14].
Simultaneously, your **SSD1306 OLED Screen** mounted on the physical terminal
updates live, displaying the calculated **Net Energy recovery ( $E_{rec}$ )**
in kWh and displaying the coordinates of the closest dispatched Sri Lankan
Waste-to-Energy processing facility (such as the *Kerawalapitiya WTE Plant* or
*Karadiyana pyrolysis pilot sites*) [cite: 12].

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#### **Stage C: Active Storage Monitoring & "Bin Full" Lockout**
* An **HC-SR04 Ultrasonic Sensor** is mounted looking down into the storage
bays [cite: 128].
* If the discarded material fills a bin and breaches your safety threshold
(distance  $< 5 \text{ cm}$ ), a **"Bin Full" event** is triggered [cite: 129].
* **Local Action:** The ESP32 flashes the **Red LED** [cite: 128], sounds a
continuous warning tone via the **Buzzer** [cite: 128], and locks the **OLED
Screen** with the text: `SYSTEM LOCKOUT: DISPATCH REQUIRED`.
* **Software Action:** The ESP32 flags the FastAPI backend [cite: 129], which
programmatically disables further weight ingestions [cite: 129]. Simultaneously,
your web app **instantly compiles and downloads the End-of-Cycle Tonnage
Manifest & Disposal Invoice PDF** (using ReportLab) to audit and verify a 100%
landfill diversion rate [cite: 14, 130].

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2. Fully Updated IoT Hardware List (Bill of Materials)

This complete list includes the **TCS34725 sensor**, the **routing gate servo**, and the **high-torque crushing servo**, keeping the budget under **Rs. 4,500** for all newly acquired parts:

Item No.	Hardware Component	Source / Status	Technical Purpose in Your Integrated Prototype	Estimated Price (LKR)
16	ESP32 DevKit	In Group Inventory	System controller; runs edge sensor loops, drives I2C, and coordinates Wi-Fi communication	0.00
18	Load Cell (5kg or 10kg)	In Group Inventory	Measures material weight (M_i) to calculate scrap valuations and pyrolysis yields	0.00
19	HX711 Amplifier Module	In Group Inventory		

128] | Amplifies and converts the analog signals from the load cell into 24-bit digital scale values [cite: 128]. | **LKR 0.00** |
NEW	**TCS34725 RGB Color & Reflectivity Sensor**	**To Buy (Local SL Shop)**	Positioned looking down at the scale. Detects object placement (Clear/Lux drop) and verifies reddish Copper [cite: 128].	**Rs. 1,000.00**
NEW	**SSD1306 OLED Screen (128x64 I2C)**	**To Buy (Local SL Shop)**	Displays real-time calculations: forecasted metal prices, calculated energy recovery ($\backslash\backslash(E_{rec})$), and target pyrolysis plants.	**Rs. 850.00**
6	**HC-SR04 Ultrasonic Sensor** [cite: 128]	*In Group Inventory* [cite: 128]	Monitors storage bin levels to trigger the **Bin Full** automatic software lockout and manifest downloads [cite: 128, 129].	**LKR 0.00**
NEW	**SG90 Micro Servo Motor**	**To Buy (Local SL Shop)**	**The Routing Gate:** Swings left, right, or center to physically direct sorted scrap into its respective bin [cite: 46].	**Rs. 750.00**
NEW	**MG996R High-Torque Servo Motor**	**To Buy (Local SL Shop)**	**The Metal Crusher:** Simulates the metal compaction press by physically rotating a crushing bar on a "HOLD" decision [cite: 46].	**Rs. 1,400.00**
28	**LED Set (Red, Yellow, Green)** [cite: 128]	*In Group Inventory* [cite: 128]	Visual alerts representing: Green = SELL, Yellow = HOLD/Crushing, Red = BIN FULL [cite: 128, 129].	**LKR 0.00**
26	**Active Buzzer** [cite: 128]	*In Group Inventory* [cite: 128]	Emits short beeps for successful scans and alarms when the storage bin is full [cite: 128, 129].	**LKR 0.00**
31 / 32	**Breadboard & Jumper Wires**	*In Group Inventory*	Provides a solderless setup to wire components to the ESP32 pins.	**LKR 0.00**
33 - 35	**Resistors, Power & USB Cables**	*In Group Inventory*	Standard electrical components and programming connectors.	**LKR 0.00**
TOTAL	-	-	**A highly advanced, panel-ready research prototype!**	
 Rs. 4,000.00 (approx. \$12.00) |

3. Dual-Protocol Pin Connection Guide (ESP32 DevKit)

To keep the wiring clean, the **TCS34725 RGB Sensor** and the **SSD1306 OLED Display** are connected in parallel to the same I2C bus pins (GPIO 21 and 22), saving ESP32 pins for your servos and sensors:

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	+-----+ ESP32 DEV KIT +-----+		
	3V3	VIN	<--- 5V Power Input (For Servos &
HC-SR04)	GND	GND	<--- Common Ground Rail
(TCS34725 & OLED SDA)	<--- GPIO 21	GPIO 5	---> (HC-SR04 Trigger)
(TCS34725 & OLED SCL)	<--- GPIO 22	GPIO 18	<--- (HC-SR04 Echo)
(HX711 DOUT)	<--- GPIO 27	GPIO 13	---> (Active Buzzer)
(HX711 SCK)	<--- GPIO 14	GPIO 12	---> (SG90 Routing Gate Servo PWM)
(Green Status)	<--- GPIO 19	GPIO 15	---> (MG996R Crusher Servo PWM)
(Yellow Status)	<--- GPIO 23	GPIO 4	---> (Red Bin-Full Warning LED)
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